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Inventor(s) McDonald; Michael B.

Percutaneous Cardiac Tissue Cutter

Abstract

A cardiac cutter for cutting cardiac valve leaflets provides a main catheter adapted for femoral insertion into a patient's vasculature in an over-the-wire method. The main catheter is deployable within a patient's heart and has a proximal bridge tube and a distal bridge tube with a cutting wire extending between them, the proximal bridge tube, distal bridge tube, and cutting wire together forming a bridge that extends outwardly from the main catheter. The cutting wire is configured to cut cardiac valve leaflets using a reciprocal motion or rotating motion, and is operable by a controller. The cutting wire is threaded through a foot positioned between the proximal bridge tube and the distal bridge tube. The foot aids in the positioning of the cutting wire against a leaflet to be cut. A commissure guide tube extends from the main catheter opposite from the cutting wire and serves to stabilize the cutter within the valve.

Inventors: **McDonald; Michael B. (Cordova, TN)**

Applicant: **McDonald; Michael B.** (Cordova, TN)

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Background/Summary

REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 63/648,960, entitled Percutaneous Mitral Valve Saw" and filed on May 17, 2024, which is fully incorporated herein by reference. This application is also a continuation-in-part of, and claims priority to, U.S. Non-Provisional application Ser. No. 18/511,054, titled "Percutaneous Cardiac Valve Cutter," filed Nov. 16, 2023, which claimed priority to U.S. Provisional Patent Application Ser. No. 63/425,708, titled "Percutaneous Cardiac Valve Saw," filed on Nov. 16, 2022. All of the foregoing are incorporated herein by reference in their entireties.

BACKGROUND AND SUMMARY
[0002] In the emerging field of structural cardiology, there is often the need for cardiac leaflet modification. A cardiac leaflet modification may be done on a previously placed bioprosthetic valve or the native valve. Making cuts in the valve leaflets can mitigate problems created by the leaflets of the first valve. When placing a second valve inside the first valve, the leaflets of the first valve will be pinned up against the valve frames creating a tube of tissue between the two valve frames. The tissue tube formed by the first valve's leaflets is often referred to as the neoskirt. Making cuts in the leaflets of the original valve with a cutter according to the present disclosure will help prevent creating an intact tube of tissue between the two valves. An intact neoskirt can limit blood flow to the coronary arteries and limit coronary catheter access to the arteries.
[0003] A cardiac cutter according to the present disclosure is provided for cutting cardiac valve leaflets via a main catheter adapted for femoral insertion into a patient's vasculature in an over-the-wire method. The main catheter is deployable within a patient's heart and has a proximal bridge tube and a distal bridge tube with a cutting wire extending between them, the proximal bridge tube, distal bridge tube, and cutting wire together forming a bridge that extends outwardly from the main catheter. The cutting wire is configured to cut cardiac valve leaflets using a reciprocal motion or rotating motion and is operable by a controller. The cutting wire is threaded through a foot positioned between the proximal bridge tube and the distal bridge tube. The foot aids in the positioning of the cutting wire against a leaflet to be cut and prevents the cutting wire from cutting tissue not within the boundaries of an inner loop of the foot. A commissure guide tube extends from the main catheter opposite from the cutting wire and serves to stabilize the cutter within the valve.
[0004] For purposes of summarizing the invention, certain aspects, advantages, and novel features of the invention have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any one particular embodiment of the invention. Thus, the invention may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Description

DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1** depicts a cutter according to an embodiment of the present disclosure.

[0006] FIG. **2** depicts a side plan view of the cutter of FIG. **1**.
[0007] FIG. **3** depicts the cutter of FIG. **1** that has been advanced into a patient's aortic valve and deployed.
[0008] FIG. **4** depicts the cutter of FIG. **3**, where the cutting wire has begun to cut into a leaflet of a patient's aortic valve.
[0009] FIG. **5** depicts the cutter of FIG. **3**, where the cutting wire has completed cutting into a leaflet of a patient's aortic valve.
[0010] FIG. **6** is a cross-sectional view of the cutter of FIG. **3** within a patient's heart, taken along section lines A-A of FIG. **3**.
[0011] FIG. **7** is a cross-sectional view of the cutter of FIG. **4** within a patient's heart, taken along section lines B-B of FIG. **4**.
[0012] FIG. **8** is a cross-sectional view of the cutter of FIG. **5** within a patient's heart, taken along section lines C-C of FIG. **5**.
[0013] FIG. **9** depicts a controller to an embodiment of the present disclosure.
[0014] FIG. **10** illustrates the main catheter of the cutter after it has been positioned in a patient's heart for cutting the leaflets of a bioprosthetic valve.
[0015] FIG. **11** depicts the cutter after it has been advanced into position in the bioprosthetic valve of FIG. **10** and the sheath has been retracted.
[0016] FIG. **12** depicts the cutter of FIG. **11** after the commissure guidewire has been deployed.
[0017] FIG. **13** depicts the cutter of FIG. **12** after the bridge has been partially deployed and
[0018] the leaflet partially cut.
[0019] FIG. **14** depicts the cutter of FIG. **13** after the bridge has been fully deployed and the leaflet fully cut.
[0020] FIG. **15** depicts the cutter of FIG. **14** after the bridge and commissure guidewire are undeployed following the cutting of the leaflet.
[0021] FIG. **16** depicts the cutter of FIG. **15** after the cutter has been partially withdrawn from the valve to reposition the cutter.
[0022] FIG. **17** depicts an alternative embodiment of a bridge of a cutter according to the present disclosure.
[0023] FIG. **18** depicts an enlarged view of the distal bridge tube and the distal bridge support, taken along detail line D of FIG. **17**.
[0024] FIG. **19** depicts an alternative embodiment of a mitral valve version of a cutter according to the present disclosure.
[0025] FIG. **20** depicts an end view of the cutter of FIG. **19**.
[0026] FIG. **21** depicts the cutter of FIG. **19** in an undeployed configuration.

DETAILED DESCRIPTION

[0027] FIG. **1** is a perspective view of a cutter **100** according to an exemplary embodiment of the present disclosure, shown in a deployed configuration. In this embodiment, the cutter **100** comprises a main catheter **103** that houses a central wire **120** for directing the cutter **100** through a patient's blood vessels (not shown) and across a cardiac valve (not shown) that is to be cut. The main catheter **103** is adapted for femoral insertion into a patient's vasculature in an over-the-wire method.
[0028] The main catheter **103** houses a cutting wire **101** that extends outwardly from the main catheter **103** when deployed. In this regard, the cutting wire **101** extends between a proximal bridge tube **102** and a distal bridge tube **104**. The proximal bridge tube **102** and the distal bridge tube **104** are both formed from hollow tubes (formed from nitinol in one embodiment), and the cutting wire **101** is threaded through the proximal bridge tube **102** and the distal bridge tube **104** such that the cutting wire **101** extends between them. The proximal bridge tube **102** and the distal bridge tube **104** project the cutting wire **101** out from the body of the main catheter **103** for the cutting operation. In the deployed configuration, the cutting wire **101** extends generally parallel to the shaft

of the main catheter **103**.

[0029] A proximal bridge support **110** is affixed to and supports the proximal bridge tube **102** when the cutter **100** is in a deployed configuration. A distal bridge support **111** is affixed to and supports the distal bridge tube **104** when the cutter is in the deployed configuration. In this regard, the proximal bridge support **110** and the distal bridge support **111** each extend from the main catheter **103** from their lower ends when the cutter **100** is in the deployed configuration, and their upper ends are affixed to and support the proximal bridge tube **102** and distal bridge tube **104**, respectively. In this manner the proximal bridge support **110** and distal bridge support **111** assist in deploying the cutting wire **101** such that the cutting wire **101**, the proximal bridge tube **102** and distal bridge tube **104** form a bridge **109** extending from the main catheter **103**. The proximal bridge support **110** and the distal bridge support **111** are formed from flexible nitinol tubing in one embodiment.

[0030] The cutting wire **101** further passes through the main catheter **103** to a controller (not shown) in the illustrated embodiment. Operation by the user of the controller controls the motion and tension of the cutting wire **101**, as discussed herein with respect to FIG. **9**.

[0031] A foot **106** extends from the main catheter **103** when the cutter **100** is in the deployed configuration. The foot **106** comprises an inner loop **107** and an outer loop **119** in the illustrated embodiment. The cutting wire **101** is threaded through the inner loop **107** of the foot **106**, such that the foot **106** constrains the cutting wire **101** within the inner loop **107**, while allowing the cutting wire **101** to freely move back and forth in a reciprocal fashion, or to rotate, while also preventing the cutting wire **101** from cutting into tissue that is not within the inner loop **107**. The inner loop **107** is approximately 0.1 cm wide in one embodiment. The foot **106** helps to direct the cutting wire **101** during the cutting operation, as further discussed herein. The outer loop **119** surrounds the inner loop **107**, adding more surface area to the foot and helping to center the foot on a leaflet (not shown) and stabilize the leaflet during the cutting operation. The outer loop **119** and the inner loop **107** are formed from flexible nitinol tubing in one embodiment.

[0032] A commissure guidewire **108** extends from the main catheter **103** when the commissure guidewire **108** is in the deployed configuration. The commissure guidewire **108** is disposed on an opposite side from (i.e., 180 degrees around the main catheter **103** from) the cutting wire **101**. The commissure guidewire **108** serves to help position the cutting wire **101** during the cutting operation, as further discussed herein. The commissure guidewire **108** is formed from flexible nitinol wire in one embodiment, and comprises irregular steps **112***a*, **112***b*, and **112***c* in the illustrated embodiment. The steps **112***a*, **112***b*, and **112***c* are bends in the commissure guidewire **108** that may aid in stabilizing the cutting wire **101** during the cutting operation.

[0033] The central catheter **103** comprises an outer sheath **105** that extends to enclose the cutting wire **101**, proximal bridge tube **102**, distal bridge tube **104**, foot **106**, the proximal bridge support **110**, distal bridge support **111**, and commissure guidewire **108** when the catheter is being passed into a patient's body to its heart through vasculature. In this regard, the sheath **105** extends to contact a nosecone **117** of the cutter **100** when the cutter **100** is in an undeployed configuration. The outer sheath **105** is retractable to reveal the cutting wire **101** and other deployable components when the catheter **103** has been placed in position in the patient's heart, as further discussed herein. The nosecone **117** comprises a nosecone opening **116**, the nosecone opening receiving the central wire **120** that the catheter **103** passes over.

[0034] As discussed herein, the proximal bridge tube **102**, the distal bridge tube **104**, the proximal bridge support **110**, and distal bridge support **111** are formed from nitinol tubing in one embodiment. The nitinol tubing is flexible and straightens when undeployed and deploys to preset shapes as shown in the illustrated configuration when the bridge **109** is deployed.

[0035] In one embodiment, the foot **106** is formed from solid nitinol wire that is flexible and straightens when constrained by the sheath **105** and deploys to the preset shape as shown in the illustrated configuration when the sheath **105** is retracted.

[0036] The cutting wire **101** comprises a diamond cutting wire in one embodiment. The motion of the cutting wire **101** may be in a one-way fashion (rotating) or two-way fashion (back and forth in a sawing motion). In one embodiment, the ends (not shown) of the cutting wire **101** feed back through the main catheter **103** to a control handle of the controller, as further discussed below. In the illustrated embodiment, the distal bridge tube **104** is affixed to the main catheter **103** at its lower end at a connection point **114**. The tubing that comprises the distal bridge tube **104** then is redirected at a turnaround point **114** and fed back into the main catheter at an opening **115** for feeding the tubing containing the cutting wire **101** back to the controller (not shown). Other embodiments may use other configurations to connect the distal bridge tube **104** to the main catheter **103** and other ways to return the tubing containing the cutting wire **101** back to the controller.
[0037] FIG. **2** is a side plan view of the cutter **100** of FIG. **1**. The proximal bridge tube **102** has an opening (not shown) in its upper end **201** through which the cutting wire **101** exits, and the distal bridge tube **104** has an opening (not shown) in its upper end **202** through which the cutting wire **101** enters. The open end **201** of the proximal bridge tube **102** faces the open end **202** of the distal bridge tube **104** when the cutter **100** is in the deployed position, such that the cutting wire extends between and moves within the proximal bridge tube **102** and the distal bridge tube **104**.
[0038] When the main catheter **103** is in a deployed configuration, the proximal bridge tube **102** extends from the main catheter **103** outwardly in a radial direction and distally towards the distal end of the cutter **100**. The distal bridge tube **104** extends from the main catheter **103** outwardly in a radial direction and forwardly towards the proximal end of the cutter **100**. The proximal bridge tube **102** and distal bridge tube **104** are in the same radial plane as one another, and both are in the same general plane a center of the foot **106**.
[0039] FIG. **3** is a cross-sectional depiction of a patient's heart **305**, where the cutter **100** of FIG. **1** has been advanced into the aortic valve **304** and deployed to cut a leaflet **303** of a stenosed aortic valve. To use the cutter **100**, the cutter **100** is advanced transfemorally through the patient's aorta **311** and across the aortic valve **304**, such that after the sheath (not shown) is retracted and the bridge **109** is partially deployed, the distal bridge tube **104** and distal bridge support **111** have entered the left ventricle cavity **310**. The cutter **100** is positioned such that the foot **106** is adjacent to or contacts the leaflet **303** desired to be cut, and the proximal bridge tube **102** is on the aortic side of the valve **304**. In this orientation, the cutting wire **101** extends from the aorta **311**, through the aortic valve **304**, and into the left ventricular cavity **310**. (Note that the bridge **109** in this illustration has been partially, but not fully deployed. FIG. **5** depicts the bridge **109** in a fully-deployed configuration.)
[0040] The foot **106** may contact, or be adjacent to, the inner wall of the aortic valve **304**. The commissure guidewire **108** may rest in the commissure (not shown) separating the leaflets adjacent to leaflet **303**, as further depicted in FIG. **6**. (One such adjacent leaflet **302** is depicted in FIG. **3**; see FIG. **6** for a top view of the valve **304** depicting all three leaflets of a valve.) The commissure guidewire **108** serves to stabilize the cutter **100** and helps to center the cutter **100**.
[0041] As illustrated in FIG. **3**, the cutting wire **101** is within the inner loop **107** of the foot **106**. In this position with little tension applied to the cutting wire **101**, and before the cutting wire motion has begun, the wire does not begin cutting into the leaflet.
[0042] FIG. **4** depicts the cutter **100** of FIG. **3**, where tension has been applied to the cutting wire **101**, and the cutting wire **101** has been moved up and down in a reciprocating motion, such that the cutting wire **101** has begun to slice into the leaflet **303**. In this view the leaflet **303** has been cut about halfway through. In this embodiment, a reciprocating sawing motion of the cutting wire **101** has been used to cause the cutting wire **101** to cut into the leaflet **303**. As discussed herein, other embodiments may employ a rotating cutting wire.
[0043] FIG. **5** depicts the cutter of FIG. **3**, where the bridge **109** has been fully deployed and the cutting wire **101** has fully cut the leaflet **303**. When the bridge was **109** only partially deployed, as

illustrated in FIGS. **3** and **4**, the cutting wire **101** could not cut fully through the leaflet **303**. But when the bridge **109** is fully deployed as shown, the cutting wire **101** can slice through the leaflet **303**, close to the wall of the valve but constrained within the foot **106** such that it cannot cut too far.

[0044] FIG. **6** is a cross-sectional view of the cutter **100** of FIG. **3**, with the cutter **100** in a patient's aortic valve **304** as depicted in FIG. **3**, taken along section lines A-A of FIG. **3**. The valve **304** comprises leaflets **301**, **302**, and **303**, and commissures **604**, **605**, and **606**. In this illustration the foot **106** is pressing against or adjacent to the leaflet **303**. The distal tip of the foot **106** is adjacent to the inner wall of the aortic valve **304**. The cutting wire **101** is threaded through and retained within the inner loop **107** of the foot **106**. The foot **106** thus guides the cutting wire **101** to cut only within the perimeter of the inner loop **107** of the foot **106**.

[0045] The main catheter **103** is generally centered within the valve **304**. The commissure guidewire **108** has been deployed to rest in the commissure **604** between leaflets **301** and **302**. The commissure guidewire pressing against the commissure and/or against the inner wall of the aortic calve **304** helps to keep the main catheter **103** centered and to stabilize the cutter **100** during the cutting operation.

[0046] FIG. **7** is a cross-sectional view of the cutter **100** of FIG. **4**, taken along section lines B-B of FIG. **4**. In this view, the cutting wire **101** has cut almost halfway through the leaflet **303**.

[0047] FIG. **8** is a cross-sectional view of the cutter **100** of FIG. **5**, taken along section lines C-C of FIG. **5**. In this view, the cutting wire **101** has cut all of the way through the leaflet **303**, and has been stopped from cutting further by the inner loop **105** of the foot **106**. After one leaflet has been cut, the cutter **100** can be repositioned to cut the remaining leaflets of the valve.

[0048] After all of the leaflets have been cut, the arm tube **104** and distal tube **102** are pulled back inside of the main catheter **103** (FIG. **1**) and the sheath **105** (FIG. **1**) is advanced to return the catheter to an undeployed configuration, and the main catheter **103** may then be removed from the patient's body.

[0049] FIG. **9** depicts a controller **900** used to control the cutter **100** (FIG. **1**) according to an embodiment of the present disclosure. The controller **900** comprises a steerable control guide **902** and a main controller **901** in the illustrated embodiment. The main catheter **103** exits a distal end **910** of the steerable control guide **902**. Moving the steerable control guide **902** with respect to the main controller **901** in the direction indicated by directional arrow **904** retracts and extends the sheath **105** (FIG. **1**). In this regard, moving the steerable control guide **902** towards the main controller **901** retracts the sheath **105**, and moving the steerable control guide **902** away from the main controller **901** extends the sheath. As illustrated in FIG. **9**, the steerable control guide **902** is positioned such that the sheath is retracted.

[0050] The steerable control guide **902** comprises a steerable control actuator **903**. The steerable control actuator **903** is actuated to move the distal end of the main catheter **103** (FIG. **1**) to aid in positioning the cutter, via flexion and extension of the main catheter **103**.

[0051] A touhy borst valve actuator **905** is positioned on the main catheter **103** between the steerable control guide **902** and the main controller **901**. The touhy borst valve serves to prevent blood from leaking from around the catheter during the cutting procedure. The touhy borst valve actuator **905** is loosened before the sheath **105** is advanced or retracted, and tightened after the sheath **105** has been advanced or retracted to lock the sheath **105** in place.

[0052] The main controller **901** comprises a commissure guide control actuator **906** that when actuated, causes the commissure guidewire **108** (FIG. **1**) to deploy. In the illustrated embodiment, the commissure guide control actuator **906** is a finger-operated dial that deploys the commissure guidewire **108** more the further the dial is turned. The user can dial the commissure guide control actuator **906** partially to partially deploy the commissure guidewire **108** and can dial the actuator **906** fully to fully deploy the commissure guidewire **108**.

[0053] The main controller **901** further comprises a bridge control actuator **907** that when actuated, causes the bridge **109** (FIG. **1**) to deploy. In the illustrated embodiment, the bridge control actuator

907 is a finger-operated dial that deploys the bridge **109** more the further the dial is turned. The user can dial the bridge control actuator **907** partially to partially deploy the bridge **109** and can dial the bridge control actuator **907** fully to fully deploy the bridge **109**.

[0054] The main controller **901** further comprises a bridge lock switch **908** that locks the bridge **109** in a deployed position and prevents it from further deploying or collapsing when it is in a desired position. The bridge lock switch **908** must be unlocked in order to collapse the bridge **109**.

[0055] The main controller **901** further comprises a cutting wire activation actuator **909** that when actuated, moves the cutting wire **101** (FIG. **1**) in a reciprocal motion in the illustrated embodiment. In one embodiment, physically turning the actuator **909** causes the cutting wire **101** to move upwards and downwards in a reciprocating movement about a centimeter in each direction.

[0056] In other embodiments, other controllers or forms of control may be used to actuate the cutting wire **101**, such as a motor turning a rotating wire.

[0057] FIG. **10** illustrates the main catheter **103** of the cutter **100** after it has been positioned in a patient's heart **305** for cutting the leaflets of a bioprosthetic valve **1001**. As discussed herein, the main catheter **103** travels to the heart **105** over a central wire **120** in an over-the-wire manner. The nosecone **117** is positioned in the left ventricular cavity **310** of the heart **105** when the cutter is in the desired position to be deployed. The sheath **105** has not ben retracted yet in the illustration of FIG. **10**.

[0058] FIG. **11** depicts the cutter **100** after it has been advanced into position in the bioprosthetic valve **1001** of FIG. **10** and the sheath has been retracted. The foot **106** is positioned against the leaflet **1002** to be cut. The bridge **109** has not yet been deployed.

[0059] FIG. **12** depicts the cutter **100** of FIG. **11** after the commissure guidewire **108** has been deployed. As was discussed above with respect to FIG. **9**, the user deploys the commissure guidewire **108** by actuating the commissure guidewire control actuator **906** (FIG. **9**) on the controller **900** (FIG. **9**). When deployed, the commissure guidewire **108** rests in the commissure **1003** opposite from the leaflet **1002** to be cut.

[0060] FIG. **13** depicts the cutter **100** of FIG. **12** after the bridge **109** has been partially deployed and the leaflet **1002** partially cut. As was discussed above with respect to FIG. **9**, the user deploys the bridge **109** by actuating the bridge control actuator **907** (FIG. **9**) on the controller **900** (FIG. **9**). Before the bridge **109** can deployed, the user must first unlock the bridge lock switch **908** (FIG. **9**) as was discussed above with respect to FIG. **9**. The user cuts through the leaflet **1002** by actuating the cutting wire actuator **909** (FIG. **9**), which moves the cutting wire **101** in a reciprocal fashion in the illustrated embodiment.

[0061] FIG. **14** depicts the cutter **100** of FIG. **13** after the bridge **109** has been fully deployed and the leaflet **1002** fully cut. As was discussed above with respect to FIG. **9**, the user fully deploys the bridge **109** by further actuating the bridge control actuator **907** (FIG. **9**) on the controller **900** (FIG. **9**). With the bridge **109** fully deployed, the cutting wire **101** will cut further into the leaflet **1002** when the cutting wire actuator **909** is actuated. However, because the cutting wire **101** is constrained within the inner loop **1005** of the foot **1006**, the wire **101** cannot cut too far or cut into the valve **1001**.

[0062] FIG. **15** depicts the cutter **100** of FIG. **14** after the bridge **109** and commissure guidewire **108** are undeployed following the cutting of the leaflet **1002**.

[0063] FIG. **16** depicts the cutter **100** of FIG. **15** after the cutter **100** has been partially withdrawn from the valve **1001** to reposition the cutter. The cutter **100** may be rotated by manually rotating the steerable control guide **902** (FIG. **9**).

[0064] FIG. **17** depicts an alternative embodiment of a bridge **1709** of a cutter **1700** according to the present disclosure. The bridge **1709** comprises a proximal bridge tube **1702** and a distal bridge tube **1704** with a cutting wire **1701** extending between the proximal bridge tube **1702** and the distal bridge tube **1703**. The cutting wire **1701** extends from an opening **1716** of the proximal bridge tube **1702** and is fed through the proximal bridge tube **1702** from a controller (not shown) for operation of the cutting wire **1701** as described with reference to FIG. **9** herein. Reference number **1701***b* depicts the cutting tube within the proximal bridge tube **1702**. Similar, the cutting wire **1701** extends from an opening **1717** of the distal bridge tube **1704** and is fed through the distal bridge tube **1704** and returns from the distal bridge tube **1704** to the controller (not shown). Reference number **1701***a* depicts the cutting tube within the distal bridge tube **1704**. Note that the cutting wire **1701** is coextensive with the wires **1701***a* and **1701***b*, and the reference numbers **1701***a* and **1701***b* are provided for the purpose of illustrating the cutting wire within the distal bridge tube **1704** and proximal bridge tube **1702** respectively.

[0065] The bridge **1709** further comprises a proximal bridge support **1710** and a distal bridge support **1711**. The proximal bridge support **1710** and the distal bridge support **1711** each comprise hollow tubing (nitinol in one embodiment) with an outer diameter slightly smaller than an inner diameter of the proximal bridge tube **1702** and the distal bridge tube **1704**, such that the proximal bridge support **1710** can fit within the proximal bridge tube **1702**, and the distal bridge support **1711** can fit within the distal bridge tube **1704**, as further discussed below.

[0066] FIG. **18** depicts an enlarged view of the distal bridge tube **1704** and the distal bridge support **1711**, taken along detail line D of FIG. **17**. The end **1801** of the distal bridge support **1711** is partially inside of the opening **1717** of the distal bridge tube **1704** when the bridge **1709** (FIG. **18**) is fully deployed. Then, as the bridge **1709** is undeployed, the end **1801** of the distal bridge support **1711** enters the distal bridge tube **1704** further. There is sufficient room within the distal bridge tube **1704** for the cutting wire **1701** and the distal bridge support **1711**. The distal bridge tube **1704** receiving the distal bridge support **1711** allows the bridge to collapse such that the sheath (not shown) can slide over the collapsed bridge. In this regard, as the bridge is undeployed, the end **1801** of the distal bridge support **1711** moves in the direction indicated by directional arrow **1715** within the opening **1717** of the distal bridge tube **1704** while the opening **1717** of the distal bridge tube moves in the direction indicated by directional arrow **1714**. In this manner, the distal bridge support **1711** is received within the distal bridge tube **1704** as the bridge is undeployed. The proximal bridge tube **1702** and proximal bridge support **1710** are substantially similar to, but mirror images of, the distal bridge tube **1704** and distal bridge support **1711** and operate in a substantially similar manner.

[0067] Referring back to FIG. **17**, the collapsing of the bridge **1709** to undeploy the cutter **1700** is also aided by the proximal end of the proximal bridge tube **1702** entering the main catheter **1703** via an opening **1719** in the main catheter. The cutting wire **1701***b* within the proximal bridge tube **1702** are then fed back to the controller.

[0068] Similarly, the distal end of the distal bridge tube **1704** enters the main catheter **1703** via an opening **1718** in the main catheter. The cutting wire **1701***a* within the distal bridge tube **1704** are then fed back to the controller.

[0069] The foot **1706** in this embodiment is substantially similar to the foot **106** (FIG. **1**) as described herein. The cutting wire **1701** is threaded through the foot **1706** in a substantially similar manner that the cutting wire **101** (FIG. **1**) is threaded through the foot **106** (FIG. **1**). The commissure guidewire **1708** is substantially similar to the commissure guidewire **108** (FIG. **1**) as discussed herein.

[0070] FIG. **19** depicts an alternative embodiment of a mitral valve version of a cutter **1900** according to the present disclosure. The cutter **1900** comprises a bridge **1909** similar to the bridge **109** discussed above with respect to FIG. **1**. Because the mitral valve is oriented differently from the other heart valves, a foot **1906** in this embodiment faces away from a nosecone **1917** of the cutter **1900**, which is opposite from the direction of the foot **106** (FIG. **1**) in the embodiments discussed above. Other than this difference the foot **1906** is substantially similar to those discussed above.

[0071] A cutting wire **1901** extends between a proximal bridge tube **1902** and a distal bridge tube **1904** in the same manner as discussed above. The cutting wire **1901** threads through the foot **1906** in the manner as discussed above with respect to the other embodiments disclosed herein. The cutting wire may cut by a reciprocal motion (i.e., sawing), rotational motion, or electrocautery. A proximal bridge support **1910** and a distal bridge support **1911** support the proximal bridge tube **1902** and the distal bridge tube **1904** respectively when the bridge **1909** is in a deployed configuration.

[0072] Guidewires **1908** extend from the cutter **1900** when deployed. Although only one guidewire **1908** is illustrated in FIG. **19**, there are two actually guidewires **1908** in this embodiment, as further depicted in FIG. **20**. The guidewires serve to brace against the posterior mitral valve leaflet (not shown) when the cutter **1900** is in use, serving to stabilize the cutter and provide a forward force in the direction of the anterior mitral valve leaflet (not shown).

[0073] The cutter **1900** comprises an outer moveable sheath **1907** that generally covers a main catheter **1903** when the sheath **1907** is fully advanced. A bridge window **1929** in the sheath **1907** over the bridge **1909** allows the bridge **1909** to extend from the cutter **1900** when the bridge **1909** is deployed. Similar, a commissure window **1930** in the sheath **1907** over the commissure guidewires **1908** allows the commissure guidewires **1908** to extend from the cutter **1900** when the commissure guidewires **1908** are deployed. A opening **1916** in the nosecone **1917** receives the wire (not shown) over which the cutter **1900** passes in an over-the-wire fashion.

[0074] FIG. **20** depicts an end view of the cutter **1900** of FIG. **19**, showing the bridge **1909**, which comprises the proximal bridge tube **1902**, distal bridge tube **1904**, proximal bridge support **1910**, distal bridge support **1911**, cutting wire **1901**, and foot **1906**. The two commissure guidewires **2008** and **2009** extend from the cutter **1900** at an opposite side from the bridge **1909**, angled radially with respect to each other at an angle of approximately 30 degrees in one embodiment.

[0075] FIG. **21** depicts the cutter **1900** of FIG. **19** in an undeployed configuration. The sheath **1907** advances towards and away from the nosecone **1917** in the direction indicated by directional arrow **2122**.

[0076] In some cases of replacement of a mitral valve (not shown), the new valve will move the anterior leaflet of the mitral valve into a position where it will compromise blood flow out of the heart. This condition is referred to as left ventricular outflow obstruction. By making a cut that is central and longitudinal in the anterior mitral valve leaflet, the left ventricular outflow obstruction can be mitigated or eliminated.

[0077] This disclosure may be provided in other specific forms and embodiments without departing from the essential characteristics as described herein. The embodiments described are to be considered in all aspects as illustrative only and not restrictive in any manner.

Claims

1. A cardiac cutting device comprising: a main catheter configured to be threaded into a patient's vasculature, the main catheter comprising a retractable external sheath, the external sheath retracted when the main catheter is at a leaflet to be cut; a proximal bridge tube and a distal bridge tube retractable within the main catheter and covered by the external sheath when the main catheter is in an undeployed configuration, the proximal bridge tube and a distal bridge tube projecting from the main catheter when the main catheter is in a deployed configuration, the proximal bridge tube and a distal bridge tube projecting radially from the main catheter in a same plane when the main catheter is in the deployed configuration; and a cutting wire extending between the proximal bridge tube and a distal bridge tube, the cutting wire projecting outwardly from the main catheter when the main catheter is in the deployed configuration, the cutting wire, the proximal bridge tube and the distal bridge tube forming a bridge when the main catheter is in the deployed configuration.

2. The device of claim 1, wherein the cutting wire is operable to cut aortic valve leaflets by moving in a reciprocating motion.

3. The device of claim 1, wherein the cutting wire is operable to cut aortic valve leaflets by moving

in a rotating motion.
4. The device of claim 1, further comprising a flexible foot extending from the main catheter when the main catheter is in the deployed configuration, the foot projecting from the main catheter between the proximal bridge tube and a distal bridge tube substantially aligned with the proximal bridge tube and a distal bridge tube, the cutting wire threaded through an inner loop in the foot, the foot retractable within the main catheter when the main catheter is in the undeployed configuration.
5. The device of FIG. **4**, wherein the foot angles away from a nosecone at a distal end of the main catheter when the device is configured for cutting mitral valve tissue.
6. The device of FIG. **4**, wherein the foot angles towards a nosecone at a distal end of the main catheter when the device is configured for cutting valve leaflets other than mitral valve tissue.
7. The device of claim 1, wherein the cutting wire comprises a diamond wire.
8. The device of claim 1, wherein the proximal bridge tube comprises a proximal bridge tube opening that faces towards the distal end of the catheter and the distal bridge tube comprises a distal bridge tube opening that faces the proximal end of the catheter, the cutting wire extending between the proximal bridge tube opening and the distal bridge tube opening.
9. The device of claim 1, further comprising a proximal bridge support affixed to the proximal bridge tube, the proximal bridge support configured to support the proximal bridge tube when the cutter is in the deployed configuration.
10. The device of claim 9, further comprising a distal bridge support affixed to the distal bridge tube, the distal bridge support configured to support the distal bridge tube when the cutter is in the deployed configuration.
11. The device of claim 1, further comprising a controller for actuating the cutting wire from outside of the patient's body, the cutting wire configured to extend from the main catheter to the controller, the controller comprising a cutting wire activation actuator for reciprocating the cutting wire when the main catheter is in the deployed configuration.
12. The device of claim 11, wherein the controller further comprises a commissure guide control actuator configured to deploy and undeploy the commissure guidewire when the main catheter is in the deployed configuration.
13. The device of claim 12, wherein the controller further comprises a bridge control actuator, the bridge control actuator operable to partially or fully deploy the bridge when the main catheter is in the deployed configuration.
14. A cardiac cutting device comprising: a main catheter adapted for femoral insertion into a patient's vasculature in an over-the-wire method, the main catheter comprising a retractable external sheath, the external sheath retracted when the main catheter is in a deployed configuration; a proximal bridge tube and a distal bridge tube extending radially from the main catheter when the main catheter is in the deployed configuration, the proximal bridge tube and a distal bridge tube projecting radially from the main catheter in a same plane; a cutting wire extending between the proximal bridge tube and a distal bridge tube.
15. The device of claim 14, wherein the proximal bridge tube and a distal bridge tube are retractable within the main catheter and covered by the external sheath when the main catheter is in an undeployed configuration.
16. The device of claim 14, wherein the cutting wire is operable to cut valve leaflets by moving in a reciprocating motion.
17. The device of claim 14, further comprising a flexible foot extending from the main catheter when the main catheter is in the deployed configuration, the foot projecting from the main catheter between the proximal bridge tube and a distal bridge tube substantially aligned with the proximal bridge tube and a distal bridge tube, the cutting wire threaded through an inner loop in the foot, the foot retractable within the main catheter when the main catheter is in the undeployed configuration.
18. The device of claim 14, further comprising a proximal bridge support affixed to the proximal bridge tube, the proximal bridge support configured to support the proximal bridge tube when the

cutter is in the deployed configuration.

19. The device of claim 18, further comprising a distal bridge support affixed to the distal bridge tube, the distal bridge support configured to support the distal bridge tube when the cutter is in the deployed configuration.

20. The device of claim 14, further comprising a controller for actuating the cutting wire from outside of the patient's body, the cutting wire configured to extend from the main catheter to the controller, the controller comprising a cutting wire activation actuator for reciprocating the cutting wire when the main catheter is in the deployed configuration.